Superposition and interference

antocciphysics | TEACHER KEY | CPA

Learning target: Add signed displacements and distinguish interference from lost energy.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 4; 3 course days later.*
Upright. An ideal free-end reflection does not invert the displacement pulse.

02. Explain a worked example (worked)
Up 3 plus down 1 gives up 2 cm.

03. Look and explain (model)
First amplitudes add; second net displacement zero at perfect overlap.

04. Build the solution (practice)
+2 cm; 0 cm.

05. Use the idea (reason)
They continue with original shapes/directions; local zero displacement does not mean energy destroyed.

06. Choose a method (mixed) *Review: Unit 5, Day 7; 12 course days later.*
$\Delta U = +20$ J.

07. Exit check (exit)
+3 mm; partly cancel.

08. STRETCH: Apply it another way (stretch)
Opposite equal pairs such as$\pm 1, \pm 2, \pm 4$ cm; waves continue propagating in a linear medium.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 4; 3 course days later.*
First $\lambda=5$ m; second $\lambda=7$ m, using $\lambda=v/f$.

02. Explain a worked example (worked)
Up 3 plus down 1 gives up 2 cm.

03. Represent and interpret (model)
First amplitudes add; second net displacement zero at perfect overlap.

04. Solve and justify (practice)
2 A and 0; stable phase relationship/coherence.

05. Analyze the reasoning (reason)
They continue with original shapes/directions; local zero displacement does not mean energy destroyed.

06. Choose a method (mixed) *Review: Unit 5, Day 7; 12 course days later.*
$Q=\Delta U+W_{by}=8$ J; heat enters.

07. Exit check (exit)
-3 mm; no, energy also depends on motion/spatial gradients and surrounding wave field.

08. STRETCH: Apply it another way (stretch)
8 cm in phase; 0 cm in antiphase. Result depends on phase, not amplitude alone.